Subject: (ELC 422) Digital Signal Processing — Spec.: Level- 4, Commp. Eng
Examiners: Dr. Samir Kamal & Dr. Sara Hassan — Time: 75 Min

Number of Pages: one | Number of Questions: 2 | Attempt all quest

Model- 2

Question 1 [20 Points]

2 a- In A/D, in detail, explain the sampling process of a continuous time signals.

b- Check the periodicity of the following signals and calculate their fundamental periods:

(i) $X(n) = \cos(3n\pi)$ 2 (ii) $x_a(t) = 3\cos(5t + \pi/6)$ 2

c- For the signal $x(t) = 2\cos(200\pi t)$ to be sampled with sampling frequency F_S = 400Hz.

2 (i) Find an expression for its corresponding sequence, x(n). Then calculate its fundame

2 (ii) Find the samples values of the first cycle in x(n) then plot them at the same figure w

Question 2 [20 Points]

2 a- Prove that the DFT, X(K), of a sequence x(n) is periodic

2 b- Show that the frequency of D. T. Sig. must be rational number.

c- For the following sequence: x(n) = {1, 1, 0, 1}:

4 (i) Using the FFT algorithm find the frequency components of x(n) and check your resu x(n) using the IDFT method.

1 (ii) Using your results, verify Parseval's theorem.

1 (iii) Plot the amplitude and the phase spectrums